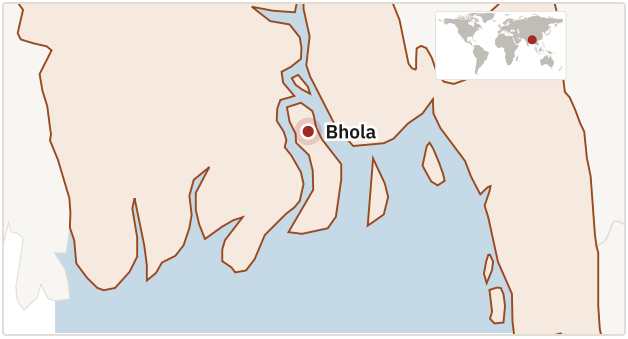
Bhola, Barisal Division

El Niño makes a winter cyclone here less likely, not impossible; what kills more people before March is infant pneumonia and diarrhoea. Stock amoxicillin, oral rehydration salts, and zinc on the chars now, and move the chars before the storm signal reaches 8.

PROVINCE	EXCESS HEAT DEATHS	WINDOW	DOMINANT HAZARD
Barisal Division (province level)	420 forecast, 220 of them in March; 390 already past (Jun–Sep)	6-month forecast, Oct 2026 – Mar 2027	Winter child pneumonia and diarrhoea; cyclones and surge; heat in March

14/14 recommendations checked. 11 corrected, 0 unverified, 5 contraindicated (the five "Do not do this" cards). Two claims were dropped in verification: a count of 78 fishers missing in Cyclone Sitrang, and a 61% shelter-use figure that the source's own table contradicts. Exposure figure: Climate Impact Lab El Niño Heat Mortality Tracker data, version of 26 October 2026, pre-peer-review, methods forthcoming. It is published for Barisal Division, not for Bhola, and it counts heat only, so it cannot see the winter deaths from pneumonia, diarrhoea, and cold that are this district's largest preventable burden before March. This briefing is at district resolution: Bhola is an island district of about 1.9 million people, so its actions are generic where the mainland and the remote chars differ. The funding target fell from the panel's BDT 13.55 million (US$109,283) to BDT 10.55 million (US$85,090) after verification (−22%), because pre-arranged cyclone money already sits unspent.

EXECUTIVE SUMMARY — AT A GLANCE



22.69°N 90.65°E, Bangladesh · Natural Earth

- **The risk.** El Niño years see about half the October–December cyclone days of La Niña years in the Bay of Bengal, but Midhili crossed near Bhola in a strong El Niño in 2023; before March the bigger killer is infant pneumonia and diarrhoea, which the heat figure does not count.
- **For you.** If a child under five breathes fast, the chest pulls in, or they cannot drink, go to the upazila health complex the same day; do not wait for a village doctor.*
- **The money.** BDT 5.55 million (US$44,760) floor, BDT 10.55 million (US$85,090) target, BDT 17.7 million (US$142,750) ceiling; crèches and child-illness supplies buy the most lives.
- **Do not.** Do not count "shelters readied" as protection: in Cyclone Remal about 2% of Bhola's people reached its 869 shelters.

For you, as a resident	9 steps	p. 2
What to do	**9** actions	p. 6
Who does what	RACI chart	p. 13
Where money does the most	funding	p. 13
Do not do this	**5** items	p. 15
What has happened here	the record	p. 17
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* Suggested formulation: check with a health worker before sharing.

FOR YOU, AS A RESIDENT 9 STEPS

For a household in Bhola, this winter's danger is less a cyclone than a child's chest infection in December or January, far from a hospital on a char with no bridge to the mainland. Cyclones remain possible until December, and the heat returns in March. The ranked actions below are written for the district administration, the civil surgeon, and their partners. These nine are the ones you can take yourself, grouped by what they protect. Prices marked as estimates are not published anywhere.

1 · Cyclones: walk to your nearest standing shelter now, check it, and agree where your cattle will go

PARTIALLY CONFIRMED FROM ACTIONS 2 AND 4

BY WHEN	COST TO YOU	PROTECTS
15 Nov 2026	None	The household and its animals

Bhola had 869 shelters and 14 raised livestock mounds (killas) in October 2024, but some near the riverbank have been lost to erosion, and in Cyclone Remal in 2024 only about 2% of Bhola's people reached a shelter. Walk the route at high tide. Check that there is a women's room, a working toilet, and water, and agree where your cattle will go, because the animals are why many families stay. On a char, leave when the Cyclone Preparedness Programme (CPP) volunteers call, not at the last signal: the boats cannot cross once the storm is close. Put your national ID card, land papers, and fisher card in a sealed plastic bag.

2 · Fishers: get sailing permission, leave your crew list and phone number at the ghat, and come back at Signal 3

PARTIALLY CONFIRMED FROM ACTIONS 3 AND 6

BY WHEN	COST TO YOU	PROTECTS
Every trip	None	The crew at sea

Since 2007, most of the people Bhola loses in storms have been on boats: in Cyclone Bulbul in 2019 a trawler sank where the Meghna meets the Ilisha, with 10 dead and 4 missing. The Marine Fisheries Act 2020 already requires sailing permission and an arrival report. Leave the skipper's number, the crew list, and your return date at the ghat so that the boat can be called back by name. Also check that your name is on the fisheries department's fisher-ID register, so that you receive the rice for the next fishing ban.

* Suggested formulation: check with a health worker before sharing.

3 · Children's chest infections: if a child breathes fast, the chest pulls in, or they cannot drink, go to the health complex the same day*

CONFIRMED FROM ACTION 1

BY WHEN	COST TO YOU	PROTECTS
December–January above all	Transport only	Babies and children under five

Pneumonia kills young children here every winter: Bhola General Hospital admitted 139 children with pneumonia in one week in January 2023. Count the breaths for a minute while the child is calm. Fifty or more a minute at 2–11 months, or 40 or more at 1–4 years, is fast breathing; so is the lower chest pulling in with each breath. Go the same day to the upazila health complex. If the child cannot drink or breastfeed, vomits everything, has a fit, or is unusually sleepy, go at once (WHO danger signs). Do not wait on a village doctor or a pharmacy.*

4 · Diarrhoea: keep oral rehydration salts and a zinc course at home for each child under five*

CONFIRMED FROM ACTION 1

BY WHEN	COST TO YOU	PROTECTS
30 Nov 2026	About BDT 100–150 (US$0.80–1.20) per child (estimate)	Children under five

Rotavirus diarrhoea peaks in winter. Oral rehydration salts (ORS) cut diarrhoea deaths in children by 69% in the studies, and an estimated 93% at full coverage (Munos et al. 2010). Give zinc for 10–14 days: 10 mg a day under six months, 20 mg a day from six months to five years. Start ORS at the first watery stool and keep breastfeeding.*

5 · Water: in the dry season, drink rainwater rather than salty tube-well or pond water, especially if you are pregnant*

CONFIRMED ROLE-SPECIFIC

BY WHEN	COST TO YOU	PROTECTS
Now, through the dry season	A covered jar (price not verified)	Pregnant women and everyone who drinks at home

* Suggested formulation: check with a health worker before sharing.

In the Khulna coast, more salt in drinking water raised the risk of (pre)eclampsia in pregnancy, most in the dry season (Khan et al., *PLoS ONE* 2014), and pregnant women who drank rainwater had systolic blood pressure 3.6–4.9 mmHg lower (Scheelbeek et al., *Hypertension* 2016). Do not assume water from a pond-sand filter is safe: in Mathbaria, cholera bacteria were found in 6 of 23 filter samples. Boil it or treat it.*

6 · Drowning: fence the pond edge and keep an adult watching children under five

PARTIALLY CONFIRMED | FROM ACTION 9

BY WHEN	COST TO YOU	PROTECTS
15 Nov 2026	About BDT 1,500 (US$12) for bamboo (estimate)	Children aged 1–4

Drowning is a leading killer of small children in Bangladesh, and Barishal Division's rate is about three times the national average. Community crèches cut drowning deaths in young children by about 80% (Anchal, *Pediatrics* 2012), but the government crèches have been closed since December 2025. Until they reopen, fence the pond or ditch nearest the house and agree which adult watches the children in the morning, when mothers are busiest.

7 · Warnings and money: save 1090, learn the CPP flags, and open a bKash account in your own name

CONFIRMED | ROLE-SPECIFIC

BY WHEN	COST TO YOU	PROTECTS
30 Nov 2026	Free; cash-out about 1.85% (BDT 92 on BDT 5,000)	Early warning and early cash

1090 is the government's free, 24-hour weather and warning line from any mobile operator. The signal numbers are wind codes for ports, so listen for what the CPP volunteers and the mosque loudspeaker tell you to do and by when. Before Cyclone Remal, WFP paid BDT 5,000 (US$40) by bKash to 30,000 households, including in Bhola, before landfall. Opening a bKash account needs only your national ID card and a SIM; open it in a woman's name if that is how the household manages money.

8 · Food: keep oil and lentils for the days the ferries stop, and buy rice only if the price is at or below October's

* Suggested formulation: check with a health worker before sharing.

PARTIALLY CONFIRMED FROM ACTION 8

BY WHEN	COST TO YOU	PROTECTS
Before 15 Nov 2026	Your usual shopping, a week ahead	The household when the island is cut off

Bhola grows more rice than it eats, but it has no bridge to the mainland, and when a storm signal stops the launches and ferries, oil, lentils, flour, and fuel cannot come in while fish and vegetables cannot leave. Keep a week of those. In past years rice prices stayed flat or rose through the aman harvest, so do not assume December is cheap: buy only if the price is at or below October's.

9 · Heat and dengue: plan for March heat without a fan, and empty standing water every week *

PARTIALLY CONFIRMED FROM ACTION 1

BY WHEN	COST TO YOU	PROTECTS
From 1 Mar 2027; dengue sweep weekly now	None	Older people, outdoor workers, and the whole household

Climate Impact Lab forecasts 220 extra heat deaths across Barisal Division in March 2027 alone, and rural power cuts run 7–10 hours a day in hot spells. Keep ORS for adults too, do outdoor work early, rest in shade at midday, and check daily on older neighbours. Barishal Division has led the country's daily dengue admissions in 2026: empty and scrub water containers, tyres, and pots every week. *

What to ask of the people in charge

- Ask the **civil surgeon's office or your upazila health complex** whether amoxicillin, ORS, and zinc are stocked at the health point nearest you, and who on the char can give them (action 1).
- Ask your **Upazila Nirbahi Officer** whether your char will be moved before Signal 8, and which boat will carry you (action 2).
- Ask the **Upazila Fisheries Officer** whether you are on the fisher-ID register, and when the ban-season rice will arrive (action 6).
- Ask the **Water Development Board** office when the breach or slump nearest you will be closed (action 7).

* Suggested formulation: check with a health worker before sharing.

WHAT THIS PLACE SHOULD DO RANKED

Ranked by expected lives saved per unit of effort before the peak. Backing lenses are shown per action; an action several lenses reached independently is the stronger bet, but the evidence behind each was checked for whether it came from somewhere like here.

1. Pre-stock amoxicillin, ORS, and zinc at all seven upazila health complexes and at the health points on the chars by 30 November, and train staff to treat child pneumonia and diarrhoea there

PARTIALLY CONFIRMED | PUBLIC HEALTH · PHILANTHROPIST | ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
Civil Surgeon, Bhola, with the upazila health complexes	30 Nov 2026	BDT 2.4 million (US$19,360) for supplies at 8 facilities, including March heat corners	Infant and child deaths from pneumonia and diarrhoea in December–January

Community case management of child pneumonia cut pneumonia deaths by 35% (95% CI 18–48%) and all-cause child deaths by 21% in a review of trials (Theodoratou et al., *International Journal of Epidemiology* 2010); a widely quoted 70% is an adjusted estimate, not a measured result. ORS cut diarrhoea deaths by 69% (Munos et al. 2010), and zinc by an estimated 23%. The need is local: Barishal's Sher-e-Bangla Medical College Hospital recorded 110 infant deaths from cold-related disease in December 2018, and Bhola's civil surgeon's office recorded 214 pneumonia and 495 diarrhoea admissions in one week of a recent winter. Do not assume the stock is there: amoxicillin dispersible tablets are not on the national essential drug list, and community-clinic drug supply has been disrupted since the health sector programme ended in June 2024. The same supplies serve heat-stroke corners in March, when the tracker puts 220 extra heat deaths in the division.

* Suggested formulation: check with a health worker before sharing.

Transferability: good; the trials are from low-income South Asian and African settings. Feasibility rests on supply, not authority, because treatment with amoxicillin is already in national child-health guidelines.

2. Move the remote chars first: agree with the district disaster committee to evacuate Dhal Char, Kukri Mukri, Char Nizam, and the outer Monpura chars at Signal 4 or 48 hours before landfall, with boats contracted in advance

PARTIALLY CONFIRMED | EMERGENCY MANAGER · FRONTLINE · HISTORIAN | ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
Deputy Commissioner and District Disaster Management Committee	Agreed by 15 Nov 2026	About BDT 900,000 (US$7,260) for three activation days (estimate), from government relief funds	People stranded on unprotected chars as the surge rises

In practice the chars are moved at Signals 8 to 10: CPP volunteers went to Kukri Mukri after Signal 8 in Cyclone Bulbul, and the mass evacuation in Remal began at Signals 9 and 10. By then boats cannot cross. In Remal, Dhal Char, Kukri Mukri, and Monpura chars flooded 4–5 feet deep and more than 25,000 people were marooned (UNB), and the Deputy Commissioner said many chars have no shelter at all. Signal 4 is our proposed local trigger, not current practice. With it go a door-by-door list of everyone who cannot leave alone (older and disabled people, pregnant women, women-headed households), each with a named neighbour, which matches the CPP's existing duty. El Niño lowers this season's odds but does not remove them: Cyclone Midhili crossed near Bhola in November 2023, during a strong El Niño.

Transferability: local. The cost is a government relief allocation, not donor money; the decision is the binding constraint.

3. Close the gaps in the existing fishing-boat system: every boat leaves its crew list and phone at the ghat, and boats are called back by name at Signal 3

PARTIALLY CONFIRMED | EMERGENCY MANAGER · FRONTLINE | ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
Upazila Fisheries Officers, with the CPP and the Coast Guard	From now; the October hilsa ban has ended	About BDT 150,000 (US$1,210) (estimate)	Fishers lost at sea, where most storm deaths now happen

* Suggested formulation: check with a health worker before sharing.

Most of the system already exists: the Marine Fisheries Act 2020 requires sailing permission and an arrival report, small boats can register online, and about 8,500 boats carry GSM trackers. What we found no written procedure for is calling untracked small boats back by name. Boats with VHF or GPS should relay the signal to those without. In Cyclone Bulbul in 2019, a trawler sank at the Meghna–Ilisha confluence off Bhola Sadar: 10 bodies were found and 4 men were missing. Inland launch ghats use a separate river signal series, so every instruction should say which series it means.

Transferability: local. The legal basis exists; the gap is small unlicensed boats.

4. Fix why people stay away from shelters: check the 100 shelters nearest the chars for a curtained women's room, lit toilets, water, and a livestock mound, announce the changes from mosques, and count who arrives at the next signal

PARTIALLY CONFIRMED | EMERGENCY MANAGER · FRONTLINE · SKEPTIC · HISTORIAN | ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
Upazila Nirbahi Officers, with Project Implementation Officers and mosque committees	30 Nov 2026	About BDT 2.0 million (US$16,130) (estimate), from shelter maintenance budgets	Deaths at home among people who were warned but did not move

The warning reaches people; leaving is the problem. In Cyclone Aila, 97% of surveyed households in Koyra were warned and about 26% evacuated (Ahsan et al. 2016). In Remal, only about 2% of Bhola's population reached its 869 shelters, which hold about 700,000. People give the same reasons each time: distance, a household head away, purdah and mixed rooms, crowding, and nowhere for the cattle. Local reporting in 2026 described Char Fasson shelters built for 600–700 people taking 2,500–3,000. The CPP had 13,860 volunteers in Bhola in October 2024 and ran a drill in Bhola Sadar in June 2026.

* Suggested formulation: check with a health worker before sharing.

Transferability: the evidence is from comparable coastal unions. The measure of success is arrivals, not shelters counted.

5. Confirm that Bhola is inside the Red Crescent's cyclone early action protocol and WFP's anticipatory cash list, and add char households to the list

PARTIALLY CONFIRMED · ECONOMIST · PHILANTHROPIST · ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
Bangladesh Red Crescent Society, Bhola unit, with WFP	15 Nov 2026	Already funded; BDT 5,000 (US$40) per household from WFP	Pre-arranged money landing elsewhere, or after the storm

The Bangladesh Red Crescent's cyclone early action protocol holds CHF 499,987 for about 150,000 people in 14 of the 19 coastal districts, approved in January 2024 for five years. It pre-activates at Signal 4 or higher, at least 72 hours ahead, with two or more models forecasting winds of 125 km/h, and activates at Signal 7 or higher, 30 hours ahead. It was not activated in 2025, and CHF 308,129 remained in February 2026. The 14 districts are not named in the protocol, and unions are chosen by impact forecast, so the Bhola unit should confirm that the district is in. Before Remal, WFP paid BDT 5,000 by bKash to 30,000 households in five districts including Bhola. The UN emergency fund's cyclone framework did not trigger for Remal; CERF gave BDT 930 million (US$7.5 million) after landfall instead.

Transferability: these are Bangladeshi instruments already in use. The action is a phone call, not a programme.

6. Pay fishers against the fisheries department's register: publish the ban-season rice lists, move the rice stuck in depots now, and press for a mobile-money top-up for the March–April 2027 ban

PARTIALLY CONFIRMED · ECONOMIST · FOOD SUPPLY · SKEPTIC · ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
Department of Fisheries, with the Upazila Nirbahi Officers and the District Food Controller	Rice moved by 30 Nov 2026; top-up decided by 15 Feb 2027	Rice already budgeted; a cash top-up is unfunded	Fishers going to sea in bad weather because the household has no food

* Suggested formulation: check with a health worker before sharing.

For the 8–29 October 2026 hilsa ban, 143,656 registered fisher families in Bhola were allocated 25 kg of rice each. For the March–April 2026 sanctuary ban, about 68,000 of 168,000 registered fishers got nothing, and there is no cash allocation. About 340 tonnes of fisher and VGD rice sat in a Tazumuddin depot in June 2026 because transport funds ran short. Reports from Char Fasson in April 2026 described party leaders submitting their own lists and genuine cardholders being dropped. The March–April ban coincides with the March heat peak.

Transferability: local. Publishing lists and moving stock need no new money; the top-up does, and no payer has been named.

7. Get the Water Development Board's list of open breaches and slumps, with closing dates before the 24 November spring tide, and enforce the removal of the Hazirganj embankment cut

PARTIALLY CONFIRMED | EMERGENCY MANAGER · SKEPTIC · HISTORIAN | ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
Bangladesh Water Development Board, Bhola	Before the spring tides of 24–27 Nov 2026	From existing emergency works; BDT 100 million (US$807,000) was spent on geobags in July 2026	Tidal flooding that needs no cyclone at all

A storm surge in May 2025 damaged about 5,200 homes, mostly in Char Fasson and Monpura. The Remal breaches of 2024, about 13 km mostly in Daulatkhan and Monpura, were repaired that year. In July 2026, about 30 metres of the concrete-block riverbank protection near the Ilisha launch terminal slumped, and the Board dropped 2,052 geobags there; a BDT 7.3 billion (US$58.9 million) project for 4 km of bank protection and 6 km of embankment awaits approval. In August 2026 the local administration confirmed that an embankment at Hazirganj had been cut for a building. As of 2020 only 10 of Char Fasson's 116 km of dyke was permanent. Spring tides fall around 9–12 and 24–27 November and 9–12 December.

* Suggested formulation: check with a health worker before sharing.

Transferability: local. This is an accountability action, not new construction.

8. Put 30 days of rice on the Manpura, Char Fasson, and Tazumuddin chars by 30 November, give food and fuel trucks priority on the Ilisha ferry when launches stop, and watch oil and lentil prices daily

PARTIALLY CONFIRMED | FOOD SUPPLY | ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
District Food Controller, with BIWTA and the Deputy Commissioner	30 Nov 2026	Transport of stock already held; protocol has no cost	Price spikes and empty shops when the island is cut off

The El Niño food signal here is weak: Bhola grows more rice than it eats, FAO expects an above-average national aman paddy harvest of 25.9 million tonnes, and national food inflation was 7.22% in September 2026, driven by fuel. The risk is the ferry: there is no bridge, launches stop on six or more routes at Signal 3, and the Ilisha ferry itself halted in depressions in May and July 2025, leaving trucks stranded. So stock already on the chars is the real hedge, starting with the rice stuck in depots. The Department of Agricultural Marketing publishes daily Barishal division prices; watch oil and lentils, which are what run short, not rice.

Transferability: local. No daily Bhola price series exists, so the division series is the trigger.

9. Press for approval of phase 2 of the government's community crèche programme with Bhola included, and open interim crèches only with the women's ministry's consent

PARTIALLY CONFIRMED | PHILANTHROPIST · PUBLIC HEALTH | ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
Ministry of Women and Children Affairs, through Bangladesh Shishu Academy	Decision by 31 Dec 2026	BDT 3.15 million (US$25,410) for 100 interim crèches, November–March	Drowning of children aged 1–4

Community crèches cut drowning deaths in young children to about a fifth (rate ratio 0.181; Anchal, *Pediatrics* 2012), at about BDT 2.11 million (US$17,008) per life saved in a later cohort study. The government's crèche project listed Bhola Sadar, Lalmohan, and Manpura in its first phase, which ended in April 2026; about 8,000 centres have been closed since December 2025, and phase 2 (BDT 8.38 billion, 30

* Suggested formulation: check with a health worker before sharing.

districts) awaits Planning Commission approval and would take 5–6 months to launch. CIPRB's current drowning programme in Barishal Division works in Bakerganj, not Bhola. Drowning peaks in the monsoon, so the winter return is lower than the annual figure suggests.

Transferability: strong; the evidence is from rural Bangladesh. Feasibility before the peak is the weak point.

* Suggested formulation: check with a health worker before sharing.

WHO DOES WHAT

RACI CHART

R = Responsible (does the work) · **A** = Accountable (one per action; signs off and answers for it) · **C** = Consulted (asked before) · **I** = Informed (told after). Built from the owners named in the verified actions. No one here has been contacted or has agreed to a role; every row is a proposal.

Action	Civil Surgeon	DC and district committee	Upazila Nirbahi Officers	CPP and Red Crescent	Fisheries Department	Coast Guard	Water Development Board	WFP	District Food Controller	Women's ministry and Shishu Academy	Mosque committees
1. Child-illness supplies and training	**A/R**		**C**	**I**							**I**
2. Move the chars first		**A**	**R**	**R**		**C**					**C**
3. Boat recall by name		**I**		**R**	**A/R**	**R**					
4. Shelters people will use		**C**	**A/R**	**C**							**R**
5. Early action protocol and WFP list		**C**		**A/R**				**R**			
6. Fishers paid against the register		**C**	**R**		**A**			**C**	**R**		
7. Breach list and Hazirganj cut		**C**	**C**				**A/R**				
8. Rice on the chars and ferry priority		**C**	**C**						**A/R**		
9. Crèches	**C**		**I**							**A/R**	

What the chart shows: accountability is spread across nine bodies, one per row, and the Deputy Commissioner is consulted on most of them. Rows 2, 5, 6, and 7 need decisions and lists more than money.

WHERE MONEY DOES THE MOST

FUNDING

The Philanthropist lens, reconciled against the ranked actions and verified. Placements are ranked by marginal lives saved per dollar. The envelope is bounded by what this place can spend before the peak, not by what it needs.

FLOOR

BDT 5.55 million (US$44,760)

TARGET

BDT 10.55 million (US$85,090)

CEILING

BDT 17.7 million (US$142,750)

LARGEST UNCERTAINTY

Whether the women's ministry will let interim crèches open while phase 2 is pending

In millions of taka: floor = crèches 3.15 + child-illness supplies 2.4 = 5.55. Target = 5.55 + CPP char kits 3.0 + a March heat contingency 2.0 = 10.55. Ceiling = 200 crèches 6.3 + 2.4 + 3.0 + a larger heat contingency 4.0 + shelter fixes 2.0 = 17.7. The target fell from the panel's 13.55 (−22%) because its BDT 5 million cyclone contingency would have duplicated the Red Crescent and CERF cyclone funds, which are unspent and

* Suggested formulation: check with a health worker before sharing.

already include an observational trigger; only a heat contingency remains. Char boats and shelter fixes are government costs and appear here only in the ceiling.

RECIPIENT	USE	AMOUNT	MECHANISM	BASIS	VERDICT
Bangladesh Shishu Academy, Bhola, with the women's ministry's consent	Action 9: restart 100 crèches, November–March	BDT 3.15 million (US$25,410)	Direct grant, paid monthly against attendance, with claw-back if phase 2 is approved	BDT 2.11 million (US$17,008) per life saved; BDT 51,600 (US$416) per crèche per year (SoLiD cohort, *Injury Epidemiology* 2021); Anchal BDT 100,700 (US$812) per DALY	Partially confirmed
Civil Surgeon, Bhola	Action 1: amoxicillin, ORS, zinc, and heat corners at 8 facilities	BDT 2.4 million (US$19,360)	Pre-positioned stock	Zinc about BDT 149,000 (US$1,200, 2001 dollars) per death averted (Robberstad 2004, via the Child Health Task Force); unit cost an estimate	Partially confirmed
CPP Bhola, through the Bangladesh Red Crescent	Actions 2 and 3: megaphones, sirens, life jackets, and batteries for 50 char units	BDT 3.0 million (US$24,200)	Pre-positioned stock	No published cost per life saved for CPP; unit cost an estimate	Unverified
Bangladesh Red Crescent, Bhola unit	Action 1: heat contingency, released on a March heat alert only	BDT 2.0 million (US$16,130)	Pooled fund	Heat is the one hazard here with no pre-arranged money	Partially confirmed
Upazila Nirbahi Officers (ceiling only)	Action 4: women's rooms, lights, and toilets in shelters	BDT 2.0 million (US$16,130)	Top-up	Overlaps government shelter maintenance; cost an estimate	Unverified

* Suggested formulation: check with a health worker before sharing.

The funder's one thing: do not add cyclone money. Pre-arranged cyclone funds already sit unspent at the Red Crescent and CERF, and El Niño makes a winter cyclone less likely. The money that buys lives here this winter is the unglamorous kind: medicine for children's chests and someone watching toddlers near the ponds.

Do not do this

CONTRAINDICATED · ADVICE THAT WOULD COST LIVES HERE

Advice that circulates for this hazard and would cost lives here, each checked against a documented case.

Do not count "shelters readied" as protection

CONTRAINDICATED

Official figures count capacity, not people. In Cyclone Remal, Bhola had 869 shelters ready, holding about 700,000 people, and only about 2% of its population used them. In Aila, 97% of surveyed households were warned and about 26% moved.

Instead: fix the reasons people stay away, and measure arrivals at the next signal (action 4).

Do not broadcast a signal number on its own

CONTRAINDICATED

The 1–11 signals describe wind at a port, and the system has not been revised in three decades; households read the numbers as their own risk and wait for a high one. Inland river ports use a separate series, which adds to the confusion. In 1970, people misread the signal.

* Suggested formulation: check with a health worker before sharing.

Instead: say what to do and by when ("go to the shelter before 4pm"), say which signal series is meant, and move the chars earlier (action 2).

Do not build new pond-sand filters or desalination plants without a paid operator, or assume filtered pond water is safe *

CONTRAINDICATED

Most public pond-sand filters in Bagerhat were not working, and some were abandoned after cyclones because no one agreed who would maintain them. In Mathbaria, cholera bacteria were found in 6 of 23 filter samples, 3 of them toxigenic, and the filter itself appeared to be a source.

Instead: repair existing filters with a named, paid caretaker, and use household rainwater tanks for new supply.

Do not distribute relief from party or union member lists

CONTRAINDICATED

Relief rice goes through union councils by rule, but in April 2026 reports from Char Fasson described party leaders submitting their own lists and genuine cardholders being dropped, and union members are accused of diverting rice. In the March–April 2026 ban, 68,000 of Bhola's 168,000 registered fishers got nothing.

Instead: distribute against the fisheries department's fisher-ID register, publish the list at the union office, and pay by mobile money where cash is used (action 6).

Do not give heat advice that depends on a fan *

CONTRAINDICATED

Rural areas lose power for 7–10 hours a day in hot spells. A fan or a cooling room that needs the grid fails when the heat is worst, and shelters crowded to several times capacity become heat traps.

* Suggested formulation: check with a health worker before sharing.

Instead: ORS for adults and children, shade, outdoor work moved to early morning, and daily checks on older people (action 1).

WHAT HAS HAPPENED HERE BEFORE — THE RECORD

YEAR	EVENT	TOLL	WHAT CHANGED AFTER
Nov 1970	The Bhola cyclone; surge at high tide under a full moon (La Niña)	300,000–500,000 dead nationally (estimates); more than 45% of Tazumuddin's roughly 167,000 people	The Cyclone Preparedness Programme was set up in its wake; the failed relief helped bring on 1971
Apr 1991	Cyclone, mostly around Chittagong (El Niño onset)	About 138,000 dead nationally	Shelter building expanded from about 400 shelters nationally
Nov 2007	Cyclone Sidr (La Niña)	About 3,400 dead nationally (estimates run higher); Bhola counts unofficial	New shelters in Char Fasson and other unions
May 2016	Cyclone Roanu (El Niño decaying)	3 dead in Bhola	—
Nov 2019	Cyclone Bulbul (weak El Niño)	A trawler sank at the Meghna–Ilisha confluence: 10 dead, 4 missing; about 50,000 people in Bhola without shelter	—
Oct 2022	Cyclone Sitrang (La Niña)	About 4 dead in Bhola; 21 fishers reported missing were later found in an Indian jail	—
Nov 2023	Cyclone Midhili crosses near Bhola (strong El Niño)	249 mm of rain in 24 hours; more than 350 houses damaged in Bhola; 7–10 dead nationally	—
May 2024	Cyclone Remal	3 dead officially (5–6 by local count); chars flooded 4–5 ft; more than 25,000 marooned; about 2% reached shelters	WFP's first anticipatory cash in Bhola, before landfall

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YEAR	EVENT	TOLL	WHAT CHANGED AFTER
May 2025	Storm surge	About 5,200 homes damaged, mostly in Char Fasson and Monpura	Breaches partly repaired
1973–2026	Meghna erosion	Manpura shrank from 148 to 114 km² (1973–2010); the district has lost a net of about 230 km² in roughly 60 years	Shelters and embankments built since the 1970s now lie in the river

The recurrence nobody has noticed: Bhola's worst cyclones, 1970 and Sidr in 2007, came in La Niña years, and El Niño years see about half as many October–December cyclone days in the Bay of Bengal. But Midhili crossed near Bhola in November 2023 in a strong El Niño. What lowered the death toll from hundreds of thousands to single figures was people and warnings, the CPP, more than concrete; and the river keeps destroying the concrete.

WHERE THE EXPERTS DISAGREE

UNRESOLVED

- **"Do not add cyclone money"** (Philanthropist) versus **"move the chars at Signal 4 with boats on contract"** (Emergency manager). Both hold: the money for evacuation already exists in government relief funds and the Red Crescent protocol, so the gap is the decision to move earlier, not funding.
- **El Niño protects Bhola this winter** (studies of October–December cyclone frequency) versus **it does not** (Midhili, 2023). El Niño cuts cyclone formation far to the south of the Bay but has little effect north of 10°N; the honest reading is "less likely, not safe".
- **Crèches save the most lives per taka** (Philanthropist) versus **they cannot open before the peak and drowning peaks in the monsoon** (verification). We kept them as the first placement, conditional on the ministry's consent.

WHAT NOBODY COVERED

BLIND SPOT

No lens costed protection from cold. The tracker counts heat only, yet winter in Barishal brings cold waves that fall hardest on infants and older people in thin houses; the child-illness action covers part of this, but warm clothing and blankets for the poorest were not assessed. Nor did any lens cover protection from violence for women in crowded shelters, which is part of why they stay away, or hypertension from salty drinking

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water among adults. And because Bhola is a district of 1.9 million people, every action here needs an upazila or union to be named before it is spent.

HOW TO READ THIS — METHOD

The panel is author-built. Eight simulated expert lenses were used to generate these recommendations. Where they agree, that is a strong hypothesis, not a consensus of the field. Every recommendation was then independently checked for evidence, transferability to this location, and feasibility before the peak; the verdict tags carry the result.

The Climate Impact Lab figure here is published for Barisal Division as a whole, at province level; it is not a district figure. The El Niño projection covers *heat only*; cyclones, surge, pneumonia, diarrhoea, cold, and drowning are additional to it and are carried here by the historical record rather than by a published number.

HOW THIS BRIEFING WAS PRODUCED — METHODOLOGY

- **Scope.** The location was given as "Bhola". It was taken as Bhola district, Barisal Division: Bhola Island and its chars, in seven upazilas (Bhola Sadar, Borhanuddin, Char Fasson, Daulatkhan, Lalmohan, Manpura, and Tazumuddin), with 1,932,518 people at the 2022 census. That is a district, not a neighbourhood, so the actions are generic at that resolution; where the risk differs, the remote chars are named. Climate Impact Lab figures are available only for Barisal Division (province level); no impact region was resolved.
- **Exposure.** Climate Impact Lab El Niño Heat Mortality Tracker data, file `v261026_adm1_output_BASE_rounded_cust.csv` (version of 26 October 2026, pre-peer-review): Barisal Division, 390 additional heat deaths June–September 2026, 420 forecast October 2026 – March 2027 (October 63, November 67, December 23, January 14, February 33, March 220), 810 in total; Bangladesh, 18,400; the world, 857,400. This update supersedes the September report *451,000 at Risk*. Method: ECMWF SEAS5 seasonal forecasts applied to temperature–mortality relationships against a 1996–2025 baseline. The figures count heat only; we inferred this from the data, because cold countries show exactly zero in every winter month. They are never added to the Impact Map's long-run projections.
- **Expert panel.** Eight simulated expert lenses (emergency manager, public health scientist, frontline responder, skeptic, economist, historian, philanthropist, and food supply chain analyst) each researched this place independently and in parallel, from web sources in English and Bengali, to a fixed brief.

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- **Reconciliation.** Duplicate actions were merged and backing lenses recorded. Convergence was tested for shared bias (whether the evidence behind an agreed action came from a setting like this one) before it counted in an action's favour. Contradictions were kept, not resolved by dropping a side. Actions were ranked by expected lives saved per unit of effort before the peak, each was given a named payer and trigger or marked unfunded, and the funding placements were reconciled against the ranked list.
- **Verification.** Six verifier agents checked every recommendation on three tests: evidence (the primary source and its effect size), transferability (whether that evidence came from somewhere like here), and feasibility (whether it can be done here before the peak). Financial actions were also checked for availability in Bangladesh, a named payer, a trigger threshold, and whether they already exist. The funding envelope was checked for real recipients, published unit costs, and duplication, then re-summed. Food actions were checked for whether each price series is actually published, and how often. A separate check tested what El Niño does to Bay of Bengal cyclones.
- **Who does what.** The RACI chart is built from the owners named in the verified actions, with one accountable party per action. It is a proposal: no organisation named in it has been contacted or has agreed to a role.
- **Verdicts.** *Confirmed*: evidence, transferability, and feasibility all hold. *Partially confirmed*: holds after the corrections shown. *Unverified*: could not be checked to a primary source. *Contraindicated*: would do harm or waste the window here.
- **Adaptation Inventory.** Not consulted. Its "Download Inventory" link leads to a registration form, and the browse page loads its data in the browser. No action here should be read as having no precedent; none was checked.
- **Money.** Costs are shown in Bangladeshi taka with US dollars in parentheses, at BDT 123.99 = US$1 (Bangladesh Bank weighted average, 29 October 2026). Figures published in Swiss francs (Red Crescent protocol) are given as published.
- **Reader.** This briefing was built at El Niño Ready's initiative, not from a request. No role was given, so the For you section is written for a resident. The "For you" steps were taken from the verified actions and from what the panel found about residents here; any step not drawn from a ranked action was verified on the same three tests. The rest of the report is written for the district administration, the civil surgeon, and their partners.
- **Health wording.** Passages where a wording error could hurt someone (children's danger signs and when to go to the health complex, oral rehydration salts and zinc doses, drinking water in pregnancy, unsafe pond water, and heat without a fan) are marked "* Suggested formulation: check with a health worker before sharing." They were checked against published sources, not by a health professional. They are suggestions, not medical advice, and anyone passing them on should confirm them with a doctor, pharmacist, nurse, or community health worker first. The marker was added on 2 November 2026.

* Suggested formulation: check with a health worker before sharing.

- **Rules.** No cost is derived by dividing a published total, and no local figure is invented; where data is thin, the report says so and shows the containing geography's number with its arithmetic. Every source is listed below with its verification status.
- **Limits.** The panel and verifiers are AI research agents working from web-available sources, not human experts, and no one in this place was interviewed. Several Bangladeshi news sites (Dhaka Tribune, ITV, Desh Rupantor, Rising BD, Jago News, Bangla Tribune) and ReliefWeb refused direct reads, so some figures rest on search excerpts. The Sidr counts for Bhola are unofficial. The spring-tide dates are calculated from the moon's phases, not taken from an official tide table. No daily Bhola price series or Bhola dengue count was found. The locator map shows Bhola at country scale; the island itself is not drawn.

References · status after verification

Climate Impact Lab, El Niño Heat Mortality Tracker data v261026 (downloaded 30 Oct 2026); tracker · confirmed

Child health: Theodoratou et al. 2010; Munos et al. 2010; Fischer Walker et al. 2010; Bhola pneumonia, Jan 2023; Bhola admissions; SBMCH, Dec 2018; zinc brief · corrected

Water: Khan et al. 2014, salinity and (pre)eclampsia; Scheelbeek et al. 2016; Mathbaria pond-sand filters; Bagerhat filters · confirmed

Drowning: Anchal, Pediatrics 2012; SoLiD cohort 2021; crèche phase 1 and 2; crèche closures; CIPRB Bhasa; Barishal drowning rate · corrected

Cyclone warning and shelters: signals explained; signal reform stalled; 869 shelters, Oct 2024; Remal shelter use; Ahsan et al. 2016; Remal, women and livestock; Bulbul, Kukri Mukri; Remal, UNB; 1090; CPP drill, June 2026 · corrected

Fishers: World Bank, marine fisheries; Bulbul trawler; Oct 2026 ban rice; March–April 2026 ban; no cash allocation; rice stuck in Tazumuddin; Char Fasson lists · corrected

Finance: Red Crescent cyclone EAP; EAP annual report 2025; WFP, Remal; OCHA anticipatory approach; Start Network, Remal; parametric insurance law · corrected

Embankments and erosion: May 2025 surge; Ilisha, July 2026; Hazirganj cut; Char Fasson dyke, 2020; Manpura area change · corrected

Food: FAO GIEWS Bangladesh; Department of Agricultural Marketing; September 2026 inflation; launch suspensions; ferry halted; rice surplus; Bhola bridge · corrected

Heat and power: rural power cuts · confirmed

ENSO and cyclones: Bhardwaj et al. 2019; INCOIS; NOAA ONI · confirmed

History: 1970 Bhola cyclone; IDS, 1970 politics; 50 years of cyclone preparedness; Roanu; Bulbul, Bhola; Sitrang fishers; Bhola District · corrected

Money: Bangladesh Bank, 29 Oct 2026 · confirmed

Exposure data · Climate Impact Lab, El Niño Heat Mortality Tracker · Impact Map · Adaptation Inventory

Generated 30 October 2026 for Bhola, Barisal Division. El Niño Ready is volunteer-run and is not affiliated with Climate Week NYC, The Climate Group, or Columbia University.

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